Research ethics and consent

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

Before you start

Recall Dataset and benchmark limitations. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

Plan time to read, investigate and explain. Keep predictions separate from observations.

Your mission

> Start with the target: Explain the mechanism and terminology of research ethics and consent. Try the recall before opening notes.

Your learning target

I can explain the **mechanism** and terminology of research ethics and **consent**.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Dataset and benchmark limitations. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Ablation studies. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Effect size and practical importance. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Explain the **mechanism** and terminology of research ethics and **consent**

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

Recall Dataset and benchmark limitations. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**

This week's end goal

Explain research ethics and **consent** using the records and boundaries of this fictional investigation.

Notice the evidence

> Investigate the supplied Research ethics and consent case. Separate observations, given facts and assumptions before choosing an explanation.

Research ethics considers purpose, burden, voluntary participation and consequences. Approval to teach does not automatically authorise a research data collection.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

> **List the given facts and one fact that is not supplied.**

STUDENT LESSON · 3 OF 18

See how it works

> Follow the mechanism for Research ethics and consent. Connect each labelled input, deciding step and result.

Use fictional records for this course. For the supplied classroom policy, a real study needs responsible review, plain-language purpose, opt-in **consent** and withdrawal handling before collection.

Do not implement real recruitment in this exercise.

Research ethics and consent — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Proposal automatically copies student essays into a study because they were submitted for grading.	Use fictional records for this course. For the supplied classroom policy, a real study needs responsible review, plain-language purpose, opt-in consent and withdrawal handling before collection. Do not implement real recruitment in this exercise.	The grading purpose does not establish research permission under the policy. Redesign around fictional data or obtain required review and voluntary permission before collection.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week’s visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

Compare the cases

> Compare the supplied cases for Research ethics and consent. Identify the detail that changes the conclusion.

Compare a familiar case with a changed case

Proposal automatically copies student essays into a study because they were submitted for grading. Show the intermediate steps and give the conclusion with its scope.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Change one thing in the pictured case

> **Change one thing**
>
> Change: A new project requests the same drawings for an excluded purpose.

> **Predicted result**
>
> **Prediction**: Do not reuse them under the existing **consent**; seek the required new authorisation.

> **Why the result follows**
>
> Reason: The new purpose falls outside the stated agreement.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

> Use the vocabulary of Research ethics and consent to explain a concrete decision, rather than only memorising definitions.

Word	Meaning
Consent	A declared voluntary permission process
Purpose limitation	Restricting use to the authorised purpose
Research burden	The cost or risk imposed on participants

Use two terms to explain a step in this week's example.

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Research ethics and consent, predict a step, then compare with its explanation.

Proposal automatically copies student essays into a study because they were submitted for grading.

The grading purpose does not establish research permission under the policy. Redesign around fictional data or obtain required review and voluntary permission before collection.

Worked explanation · Research ethics and consent

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Research ethics and consent

Consent is specific, not universal. A fictional study asks adults to submit anonymous drawings for one pattern-recognition investigation.

The facts we will use

Supplied facts

Consent covers this study only. It excludes public posting of originals and unrelated face-recognition research.

AI ACADEMY / AI-7/19

Research ethics and consent

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. State the purpose

Explain the drawing task and planned use before participation.



2. Respect exclusions

Do not publish originals or transfer them to the unrelated project.



3. Keep the agreement

Record permitted uses and the approved withdrawal procedure.

CHANGE ONE THING

A new project requests the same drawings for an excluded purpose.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. State the purpose	Explain the drawing task and planned use before participation.
2. Respect exclusions	Do not publish originals or transfer them to the unrelated project.
3. Keep the agreement	Record permitted uses and the approved withdrawal procedure.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Research ethics includes what people agree to, how they can decline and how risks are addressed. A contribution does not grant unlimited reuse.

What this example does not establish

Keep the stated limit

This fictional adult scenario is not approval for research with children; real oversight and applicable rules are separate requirements.

Keep the stated limit

Read the labelled picture: Identify which uses the drawing **consent** permits and excludes. Can the unrelated project reuse the drawings under this agreement, and what new authorisation would be needed?

Keep the stated limit

Change one thing: A new project requests the same drawings for an excluded purpose. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Research ethics and consent claim and explain a repair using the supplied evidence.

This classroom policy exercise does not determine real legal or institutional requirements.

Check the claim against the facts

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Locate the earliest unsupported step

A report omits this qualification: This classroom policy exercise does not determine real legal or institutional requirements. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

State one repair and the checks you would rerun.

STUDENT LESSON · 8 OF 18

Practise together

> Practise Research ethics and consent with support, then explain the deciding step in your own words.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Explain your prediction to a partner before comparing results.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Research ethics and **consent**”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Research ethics and **consent**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

> Identify which uses the drawing **consent** permits and excludes. Can the unrelated project reuse the drawings under this agreement, and what new authorisation would be needed?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Research ethics and consent. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

Predict, investigate and preserve the evidence

Proposal automatically copies student essays into a study because they were submitted for grading. Show the intermediate steps and give the conclusion with its scope.

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	

Record	Your evidence
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

Build your understanding

> Move from recall to tracing, diagnosis and transfer in Research ethics and consent. Explain why each step is justified.

Explain → apply → challenge

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

A report omits this qualification: This classroom policy exercise does not determine real legal or institutional requirements. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Explain which evidence supports your conclusion and which claim remains unproved.

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Reproducible AI Research Study. Keep an annotated decision and the evidence supporting it.

Add a labelled research ethics and **consent evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of research ethics and **consent**.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence for your project

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

STUDENT LESSON · 12 OF 18

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

> **Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.**

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My limited claim and a useful next check

STUDENT LESSON · 13 OF 18

Study a complete case

> Revisit the complete Research ethics and consent case for supported practice. Repeated examples do not become fresh assessments.

Complete evidence record

Research ethics and consent — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Proposal automatically copies student essays into a study because they were submitted for grading.	Use fictional records for this course. For the supplied classroom policy, a real study needs responsible review, plain-language purpose, opt-in consent and withdrawal handling before collection. Do not implement real recruitment in this exercise.	The grading purpose does not establish research permission under the policy. Redesign around fictional data or obtain required review and voluntary permission before collection.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Apply a new case independently

A fictional policy requires opt-in; proposal treats silence as agreement and grades participation. Audit.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

A fictional policy requires opt-in; proposal treats silence as agreement and grades participation. Audit.

> **My independent procedure, result and limitation**

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Research ethics and consent, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: **Consent** is specific, not universal. A fictional study asks adults to submit anonymous drawings for one pattern-recognition investigation.

By the end, I can	How I will show it
Explain the mechanism and terminology of research ethics and consent	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Research ethics and consent without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of research ethics and **consent**

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Consent is specific, not universal. A fictional study asks adults to submit anonymous drawings for one pattern-recognition investigation.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Record permitted uses and the approved withdrawal procedure.

1. State the purpose

2. Keep the agreement

3. Respect exclusions

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Keep the agreement Record permitted uses and the approved withdrawal procedure.

Research ethics includes what people agree to, how they can decline and how risks are addressed. A contribution does not grant unlimited reuse.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: A new project requests the same drawings for an excluded purpose. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. Do not reuse them under the existing **consent**; seek the required new authorisation.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Do not reuse them under the existing **consent**; seek the required new authorisation. The new purpose falls outside the stated agreement.

Repair a missing building block

Review Dataset and benchmark limitations (ai-7/18). Goal: Explain the **mechanism** and terminology of dataset and benchmark limitations. Short practice: Report the indoor benchmark with its denominator and missing condition.

After two correct outdoor cases out of four are added, what slice rate is supported and what remains uncertain?. Return to this week and explain what you changed.

Review Ablation studies (ai-7/16). Goal: Explain the **mechanism** and terminology of ablation studies. Short practice: Calculate the filter ablation's two rates and count change.

Which single component is altered, and why does also changing the model prevent isolating the filter?. Return to this week and explain what you changed.

Review Effect size and practical importance (ai-7/13). Goal: Explain the **mechanism** and terminology of effect size and practical importance. Short practice: Calculate the absolute error-rate reduction from 20/1000 to 15/1000.

Does it meet the ten-fewer-errors threshold, and what changes when B has eight errors?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

Visual concept journey

> Explain each Research ethics and consent graphic using its labels, deciding evidence and relevant limitation.

01. Research ethics and consent: the case



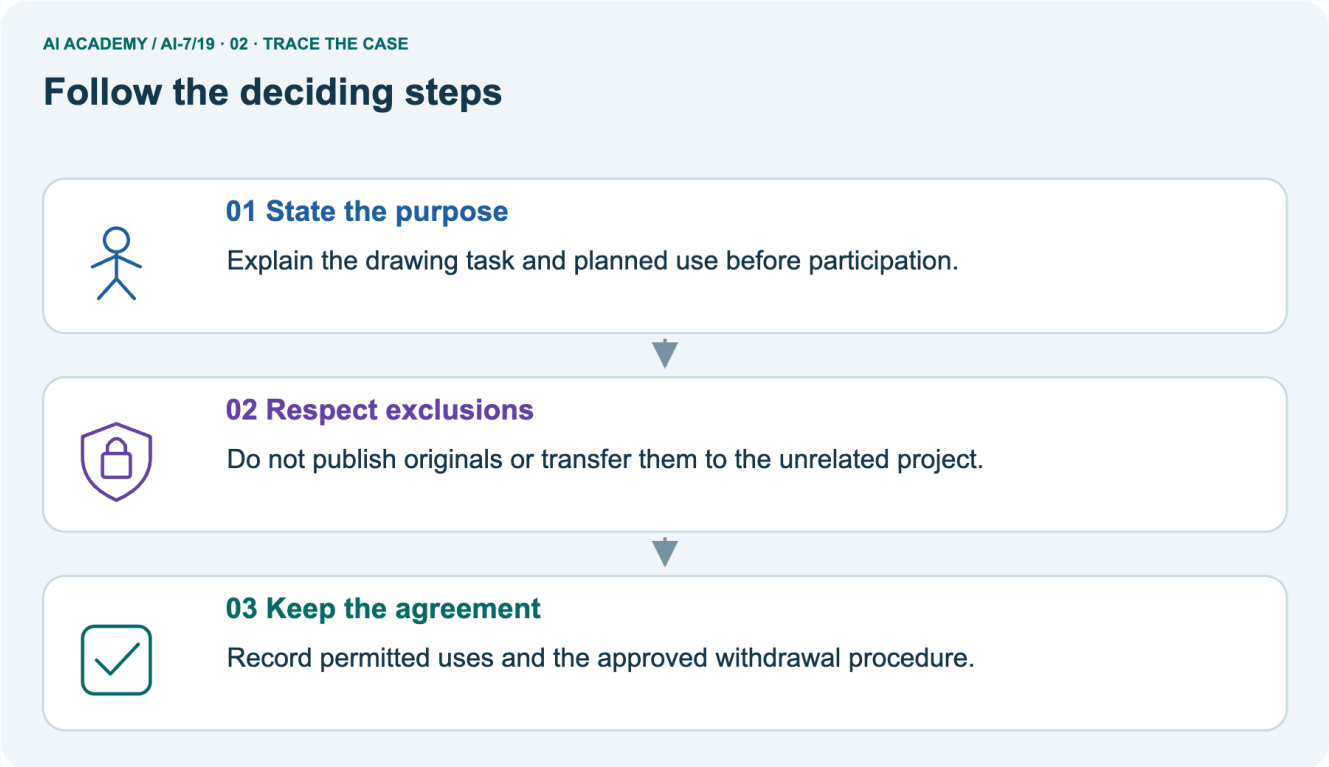
Follow the labelled picture with the matching explanation. Open the vector diagram

Consent is specific, not universal. A fictional study asks adults to submit anonymous drawings for one pattern-recognition investigation. Supplied facts: **Consent** covers this study only.

It excludes public posting of originals and unrelated face-recognition research.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

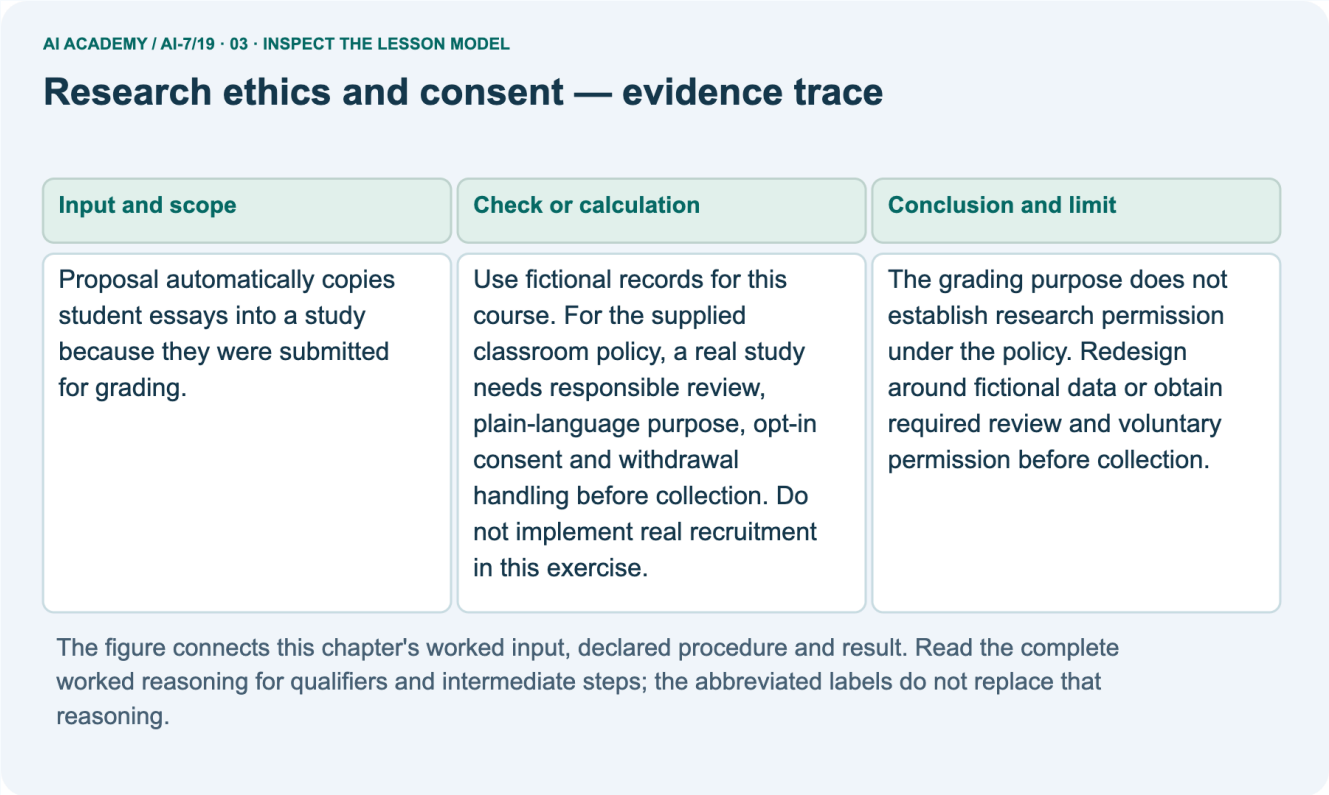


Follow the labelled picture with the matching explanation. Open the vector diagram

State the purpose: Explain the drawing task and planned use before participation. → Respect exclusions: Do not publish originals or transfer them to the unrelated project. → Keep the agreement: Record permitted uses and the approved withdrawal procedure.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Research ethics and consent — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Research ethics and **consent** — **evidence** trace. Proposal automatically copies student essays into a study because they were submitted for grading. | Use fictional records for this course.

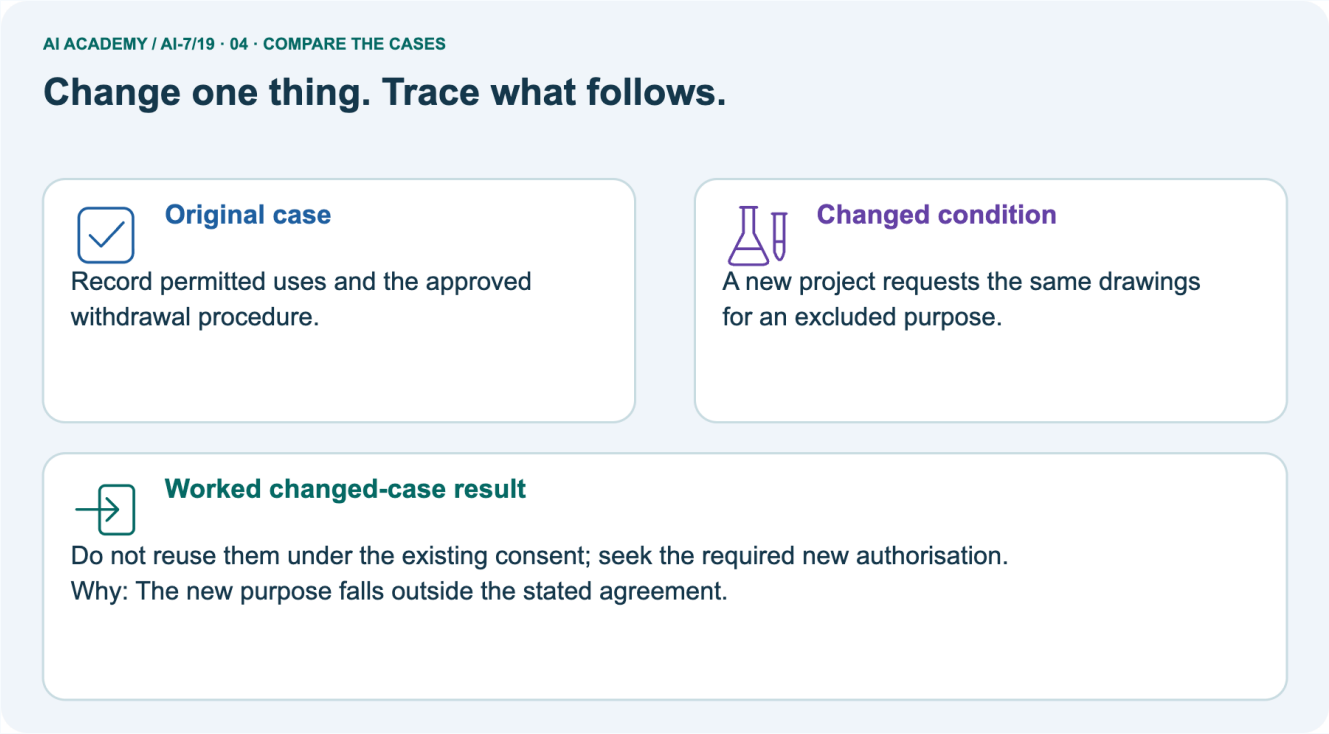
For the supplied classroom policy, a real study needs responsible review, plain-language purpose, opt-in **consent** and withdrawal handling before collection.

Do not implement real recruitment in this exercise. | The grading purpose does not establish research permission under the policy.

Redesign around fictional data or obtain required review and voluntary permission before collection.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



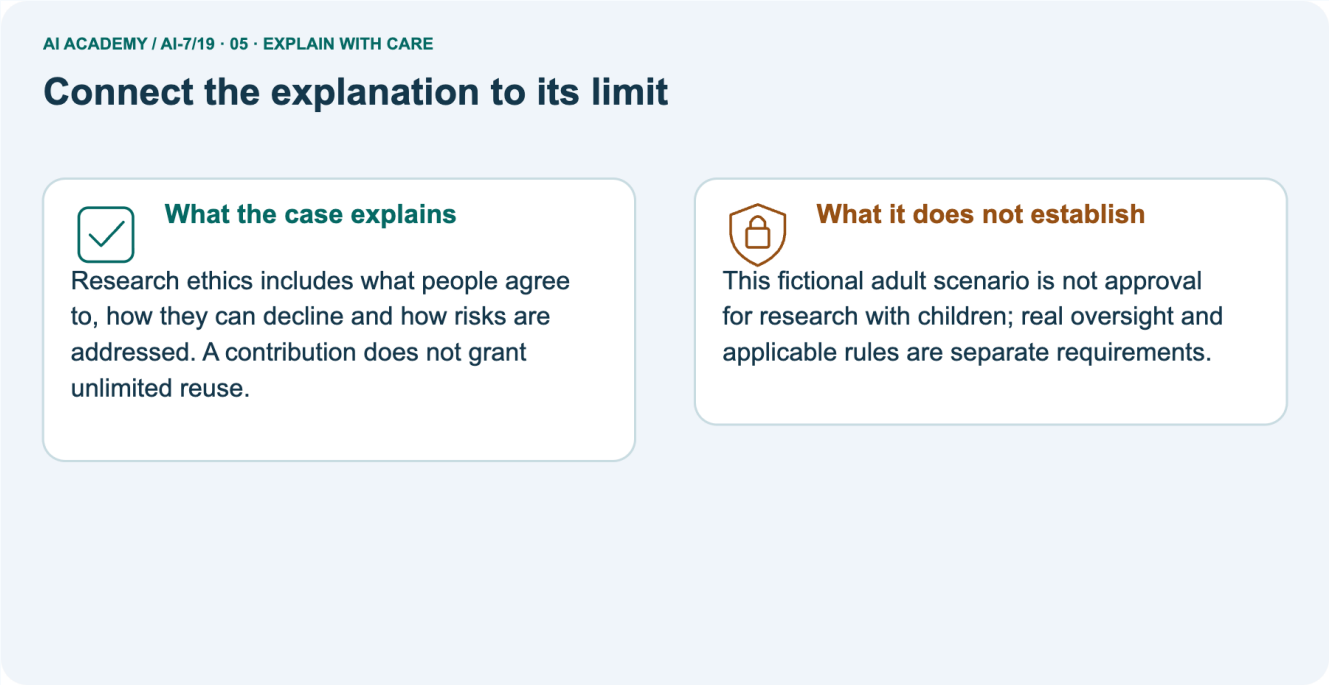
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Record permitted uses and the approved withdrawal procedure. Changed condition: A new project requests the same drawings for an excluded purpose.

Expected result: Do not reuse them under the existing **consent**; seek the required new authorisation. Reason: The new purpose falls outside the stated agreement.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



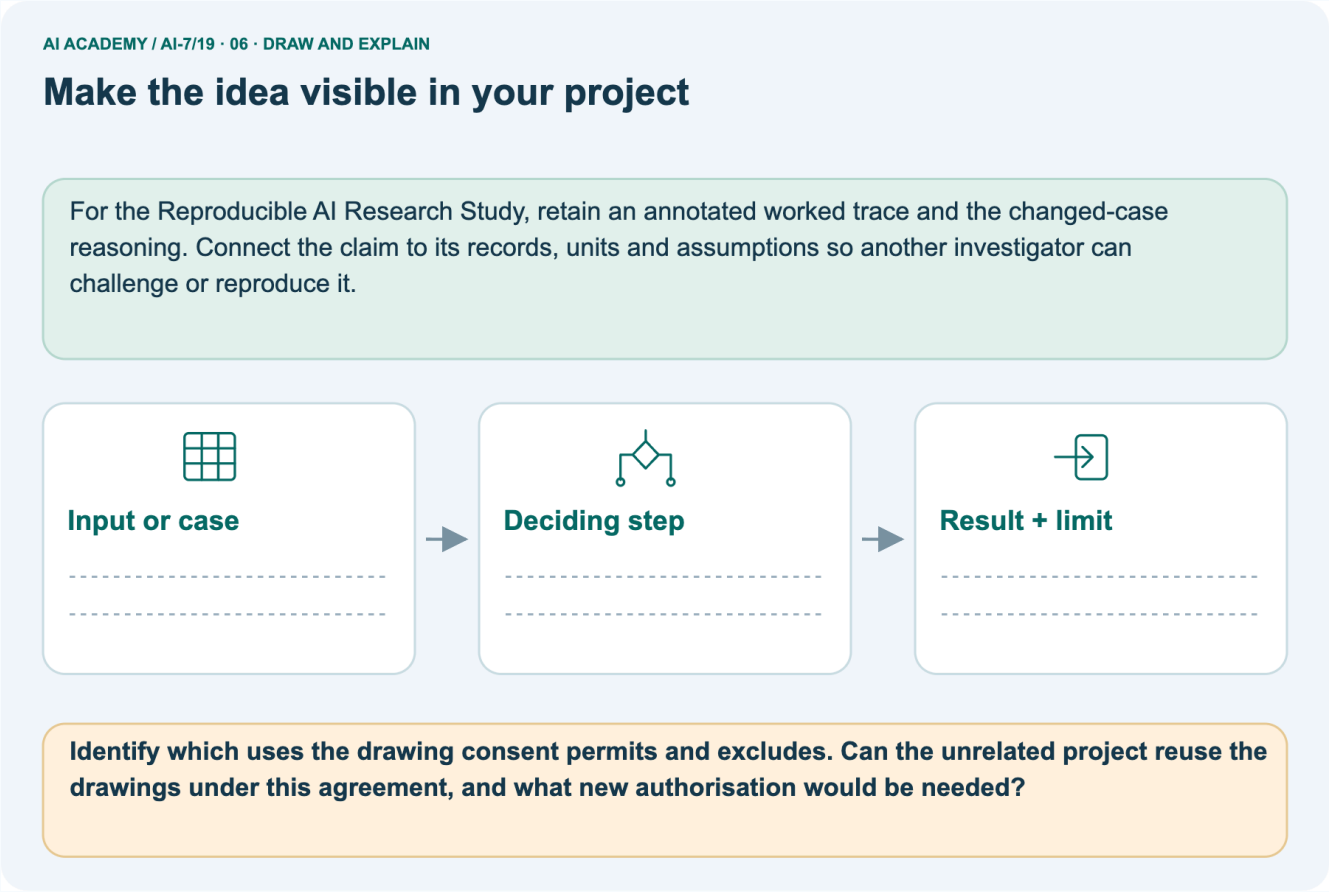
Follow the labelled picture with the matching explanation. Open the vector diagram

Research ethics includes what people agree to, how they can decline and how risks are addressed. A contribution does not grant unlimited reuse.

Limit: This fictional adult scenario is not approval for research with children; real oversight and applicable rules are separate requirements.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Identify which uses the drawing **consent** permits and excludes.

Can the unrelated project reuse the drawings under this agreement, and what new authorisation would be needed?

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.